SG No. 210 O^4 $F4_132$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `23.`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `8b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{4},\,\frac{3}{4},\,\frac{1}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `16c`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{5}{8}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{3}{8},\,\frac{5}{8},\,\frac{7}{8}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{5}{8},\,\frac{7}{8},\,\frac{3}{8}\right]$	`[4,6,11,16,18,23]`

Table 4: Wyckoff site: `16d`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{5}{8},\,\frac{5}{8},\,\frac{5}{8}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{1}{8}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{7}{8},\,\frac{1}{8},\,\frac{3}{8}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{1}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[4,6,11,16,18,23]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\,x,\,x]$	`[1,5,9]`
2	$\left[-x,\,\frac{1}{2}-x,\,x+\frac{1}{2}\right]$	`[2,7,12]`
3	$\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,-x\right]$	`[3,8,10]`
4	$\left[x+\frac{1}{2},\,-x,\,\frac{1}{2}-x\right]$	`[4,6,11]`
5	$\left[x+\frac{3}{4},\,x+\frac{1}{4},\,\frac{3}{4}-x\right]$	`[13,17,21]`
6	$\left[\frac{1}{4}-x,\,\frac{1}{4}-x,\,\frac{1}{4}-x\right]$	`[14,19,24]`
7	$\left[x+\frac{1}{4},\,\frac{3}{4}-x,\,x+\frac{3}{4}\right]$	`[15,20,22]`
8	$\left[\frac{3}{4}-x,\,x+\frac{3}{4},\,x+\frac{1}{4}\right]$	`[16,18,23]`

Table 6: Wyckoff site: `48f`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4]`
2	$[-x,\ \frac{1}{2},\ \frac{1}{2}]$	`[2,3]`
3	$[0,\ x,\ 0]$	`[5,8]`
4	$[\frac{1}{2},\ -x,\ \frac{1}{2}]$	`[6,7]`
5	$[0,\ 0,\ x]$	`[9,12]`
6	$[\frac{1}{2},\ \frac{1}{2},\ -x]$	`[10,11]`
7	$[\frac{3}{4},\ x+\frac{1}{4},\ \frac{3}{4}]$	`[13,16]`
8	$[\frac{1}{4},\ \frac{1}{4}-x,\ \frac{1}{4}]$	`[14,15]`
9	$[x+\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[17,20]`
10	$[\frac{3}{4}-x,\ \frac{3}{4},\ \frac{1}{4}]$	`[18,19]`
11	$[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}-x]$	`[21,24]`
12	$[\frac{1}{4},\ \frac{3}{4},\ x+\frac{3}{4}]$	`[22,23]`

Table 7: Wyckoff site: `48g`, site symmetry: `..2`

No.	position	mapping
1	$[\frac{1}{8},\ y,\ \frac{1}{4}-y]$	`[1,19]`
2	$[\frac{7}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y]$	`[2,17]`
3	$[\frac{3}{8},\ y+\frac{1}{2},\ y+\frac{3}{4}]$	`[3,20]`
4	$[\frac{5}{8},\ -y,\ y+\frac{1}{4}]$	`[4,18]`
5	$[\frac{1}{4}-y,\ \frac{1}{8},\ y]$	`[5,14]`
6	$[\frac{3}{4}-y,\ \frac{7}{8},\ \frac{1}{2}-y]$	`[6,16]`
7	$[y+\frac{3}{4},\ \frac{3}{8},\ y+\frac{1}{2}]$	`[7,13]`
8	$[y+\frac{1}{4},\ \frac{5}{8},\ -y]$	`[8,15]`
9	$[y,\ \frac{1}{4}-y,\ \frac{1}{8}]$	`[9,24]`
10	$[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{7}{8}]$	`[10,22]`
11	$[y+\frac{1}{2},\ y+\frac{3}{4},\ \frac{3}{8}]$	`[11,23]`
12	$[-y,\ y+\frac{1}{4},\ \frac{5}{8}]$	`[12,21]`

Table 8: Wyckoff site: `96h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}]$	`[2]`
3	$[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z]$	`[3]`
4	$[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y]$	`[6]`
7	$[-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}]$	`[7]`
8	$[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x]$	`[10]`
11	$[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x]$	`[11]`
12	$[-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}]$	`[12]`

continued ...

Table 8

No.	position	mapping
13	$\left[y+\frac{3}{4},\, x+\frac{1}{4},\, \frac{3}{4}-z\right]$	[13]
14	$\left[\frac{1}{4}-y,\, \frac{1}{4}-x,\, \frac{1}{4}-z\right]$	[14]
15	$\left[y+\frac{1}{4},\, \frac{3}{4}-x,\, z+\frac{3}{4}\right]$	[15]
16	$\left[\frac{3}{4}-y,\, x+\frac{3}{4},\, z+\frac{1}{4}\right]$	[16]
17	$\left[x+\frac{3}{4},\, z+\frac{1}{4},\, \frac{3}{4}-y\right]$	[17]
18	$\left[\frac{3}{4}-x,\, z+\frac{3}{4},\, y+\frac{1}{4}\right]$	[18]
19	$\left[\frac{1}{4}-x,\, \frac{1}{4}-z,\, \frac{1}{4}-y\right]$	[19]
20	$\left[x+\frac{1}{4},\, \frac{3}{4}-z,\, y+\frac{3}{4}\right]$	[20]
21	$\left[z+\frac{3}{4},\, y+\frac{1}{4},\, \frac{3}{4}-x\right]$	[21]
22	$\left[z+\frac{1}{4},\, \frac{3}{4}-y,\, x+\frac{3}{4}\right]$	[22]
23	$\left[\frac{3}{4}-z,\, y+\frac{3}{4},\, x+\frac{1}{4}\right]$	[23]
24	$\left[\frac{1}{4}-z,\, \frac{1}{4}-y,\, \frac{1}{4}-x\right]$	[24]